# Gravitational Wave Detection Using Deep Learning

7-Minute Presentation Speech — CS 4700 | Kayd Craig | July 2026

## Slide 1 — Title (~30 seconds)

Hey everyone. My project is about using machine learning to detect gravitational waves — the ripples in space that happen when two massive objects like black holes crash into each other. The goal was to build a system that can look at detector data and automatically figure out what it's seeing.

Tip: Make eye contact and speak slowly — the audience needs a moment to settle in.

## Slide 2 — The Problem (~45 seconds)

The way scientists currently do this is called matched filtering. Basically you have a giant library of what different signals are supposed to look like, and you compare your detector data against every single one until you find a match. It works, but it's slow and it can only find things that are already in your library. Machine learning is interesting here because instead of hand-coding what to look for, you just show the model a ton of examples and let it figure it out on its own.

Tip: Point to the two-column comparison on screen as you talk through each side.

## Slide 3 — Data Pipeline (~60 seconds)

For data I used a physics toolkit called PyCBC to generate about 15,000 fake gravitational wave signals — 5,000 black hole mergers, 5,000 neutron star mergers, and 5,000 pure noise segments. I also downloaded real data from the three confirmed detections LIGO made in its first observing run. Before any of that goes into a model, it gets cleaned up — filtered to remove frequencies we don't care about, then whitened so the noise looks the same everywhere, then normalized so everything is on the same scale.

Tip: Walk through the pipeline diagram left to right as you explain each step.

## Slide 4 — Model Architectures (~75 seconds)

I built three models. The main one is a CNN-LSTM. The CNN part scans across the signal looking for short patterns, kind of like how image recognition CNNs scan for edges and shapes. Then the LSTM takes those patterns and looks at how they change over time — which matters a lot because gravitational wave signals sweep from low to high pitch as the objects spiral in. I also tried a ResNet image classifier on spectrogram images of the signals, and a Random Forest as a simple baseline. The Random Forest just uses 18 hand-crafted statistics like how spiky the signal is and how much energy is in different frequency ranges.

Tip: This is the most technical slide — keep it moving and don't get lost in the architecture details.

## Slide 5 — Training Curves (~60 seconds)

So here's where things got interesting. The ResNet is clearly overfitting — it gets really good on training data but gets worse and worse on new data. That's a classic problem. But the CNN-LSTM is doing something even worse — the loss just sits completely flat the whole time. It's not learning anything at all. That flat number, 0.693, is actually the exact value you'd get from a model that's just randomly guessing. So something fundamental is wrong.

Tip: Pause after 'something fundamental is wrong' — let the audience feel the suspense before flipping to the next slide.

## Slide 6 — Feature Separability Diagnostic (~75 seconds)

I ran a diagnostic to figure out what. I took samples from each class, whitened them properly, and computed some basic statistics. And the result is right here on this slide — the bars are almost identical across all three classes. Black hole mergers, neutron star mergers, and pure noise all look statistically the same after whitening. Here's why. The signal strength we're working with — SNR 8 to 30 — sounds decent, but that's a cumulative number across the whole signal. At any individual moment in time, the signal is only about 0.23 times the noise level. It's completely buried. Matched filtering works because it adds up all those tiny contributions across thousands of time steps. Our models are trying to look at individual features and there's just nothing there to see.

Tip: This is your strongest slide. Pause before saying 'the bars are almost identical' and let the audience look at the chart first.

## Slide 7 — Results (~45 seconds)

So the numbers back that up. Every model lands right around 33% accuracy on the three-class problem — which is exactly what you'd expect from random guessing with three equal classes. The binary test hit 66% but that's just the model saying signal for everything — it learned nothing real. And notice the recall chart on the right — BBH gets zero. The model never once correctly identified a black hole merger.

Tip: Keep your tone confident here — a null result that you understand is better than a positive result you can't explain.

## Slide 8 — Conclusions (~30 seconds)

So to wrap up — I built the full pipeline, got real LIGO data working, ran three different models, and found a genuinely interesting result: at realistic detection strengths, the signal is so buried in noise that no simple feature can find it. That's not a failure of implementation — it's the reason matched filtering was invented in the first place. The natural next step is to feed the matched filter output directly into the neural network as an extra input. George and Huerta showed in their 2018 paper — Deep Learning for Real-time Attenuation of LIGO Noise — that this approach gets you to over 97% accuracy at these same signal strengths. Thanks, happy to take questions.

Tip: Smile when you say 'happy to take questions' — you know this material well.

Total time: ~7 minutes at a comfortable pace